```
2.5
rayyan@DESKTOP-UFNN1JS:~/xv6-labs-2026$ cat answers-traps.txt
1. The registers a0 through a7 contain function arguments. In main's call to printf, register a2 contains 13.

2. There is no explicit call to f or g in main's assembly. The compiler optimizes the function calls, calculating f(8) + 1 as 12 directly.

3. The function printf is located at address 0x6fe.

4. The value in register ra just after the call to printf is 0x34, the address of the instruction immediately after the call.

5. The output is:
   He110 World

6. On a big-endian RISC-V system, i would be 0x726c6400. The value 57616 does not need to change because it is printed as a hexadecimal number independently of the string's byte order.

7. The value printed after y= is unpredictable because only one argument is supplied for two format specifiers. printf attempts to read a second argument that was not provided, resulting in undefined behavior.
rayyan@DESKTOP-UFNN1JS:~/xv6-labs-2026$
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rayyan@DESKTOP-UFNN1JS:~/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README  user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie us
er/_logstress user/_forphan user/_dorphan user/_sync  user/_call user/_bttest user/_alarmtest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1068 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x
0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ alarmtest
test0 start
.........................................alarm!
test0 passed
test1 start
......alarm!
.....alarm!
.....alarm!
.....alarm!
.....alarm!
.....alarm!
.....alarm!
.....alarm!
......alarm!
.....alarm!
.....alarm!
test1 passed
test2 start
................................................................................alarm!
test2 passed
test3 start
test3 passed
$
```

```
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mkfs/mkfs fs.img README  user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie us
er/_logstress user/_forphan user/_dorphan user/_sync  user/_call user/_bttest user/_alarmtest
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0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ bttest
backtrace: 0x0000000080001f0c 0x0000000080001de8 0x0000000080001b16
$
```